(a) Given the image and a filter, calculate the convoluting result.

- i). Given an image of 1×8 resolution and a filter of 1×3 size as following, show the result of convoluting the image with filter

Image: $I_1 = [56 \quad 64 \quad 79 \quad 98 \quad 115 \quad 126 \quad 132 \quad 133]$
Filter: $F_1 [1 \quad 0 \quad -1]$

Answer:

Flip the filter: $[1 \quad 0 \quad -1] \rightarrow [-1 \quad 0 \quad 1]$

$I_1 * F_1$
$= [79 - 56 \; 98 - 64 \; 115 - 79 \; 126 - 98 \; 132 - 115 \; 133 - 126] = [23 \; 34 \; 36 \; 28 \; 17 \; 7]$

(a) Given the image and a filter, calculate the convoluting result.

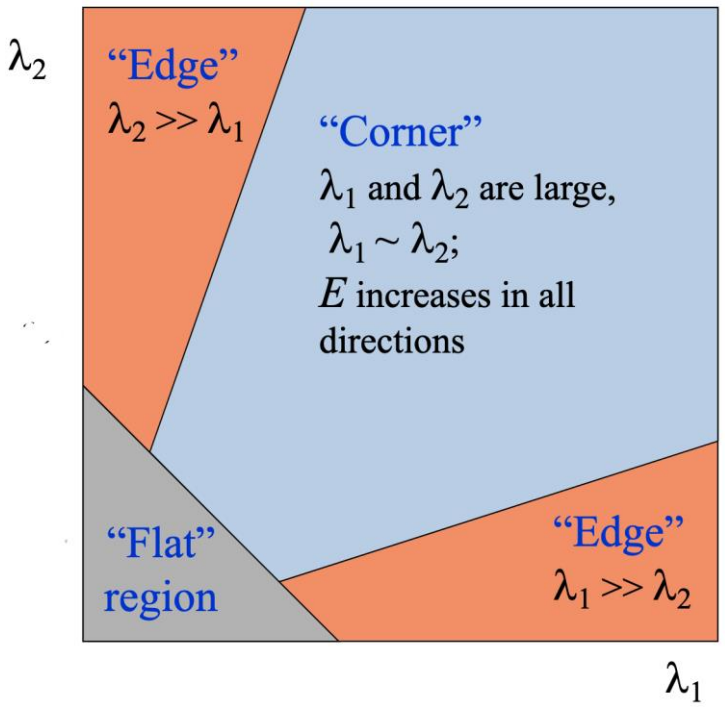
- ii). Given an image of 4x5 resolution and a filter of 3x3 size as following, show the result of convoluting the image with filter

$$I_2 = \begin{bmatrix} 4 & 6 & 7 & 5 & 8 \\ 1 & 2 & 8 & 9 & 8 \\ 7 & 7 & 5 & 9 & 9 \\ 3 & 9 & 5 & 5 & 6 \end{bmatrix} \qquad F_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Answer: $I_2 * F_2 = \begin{bmatrix} 20 & 15 & 21 \\ 16 & 24 & 19 \end{bmatrix}$

(b) Can Harris corner detector be used to detect edges? Explain your answer by describing how Harris matrix will respond to edges.

Answer: Yes, when one eigenvalue has large value and the other has small value.



(c). Explain how SIFT can achieve invariances in (i) illumination, (ii) scale, (iii) rotation.

(i). illumination : the SIFT descriptor is based on gradient magnitude, instead of pixel value.

(ii). Scale: SIFT detects the interest points across different scale levels.

(iii). Rotation: the image is firstly rotated to the dominant orientation, thus SIFT is invariant to rotation.

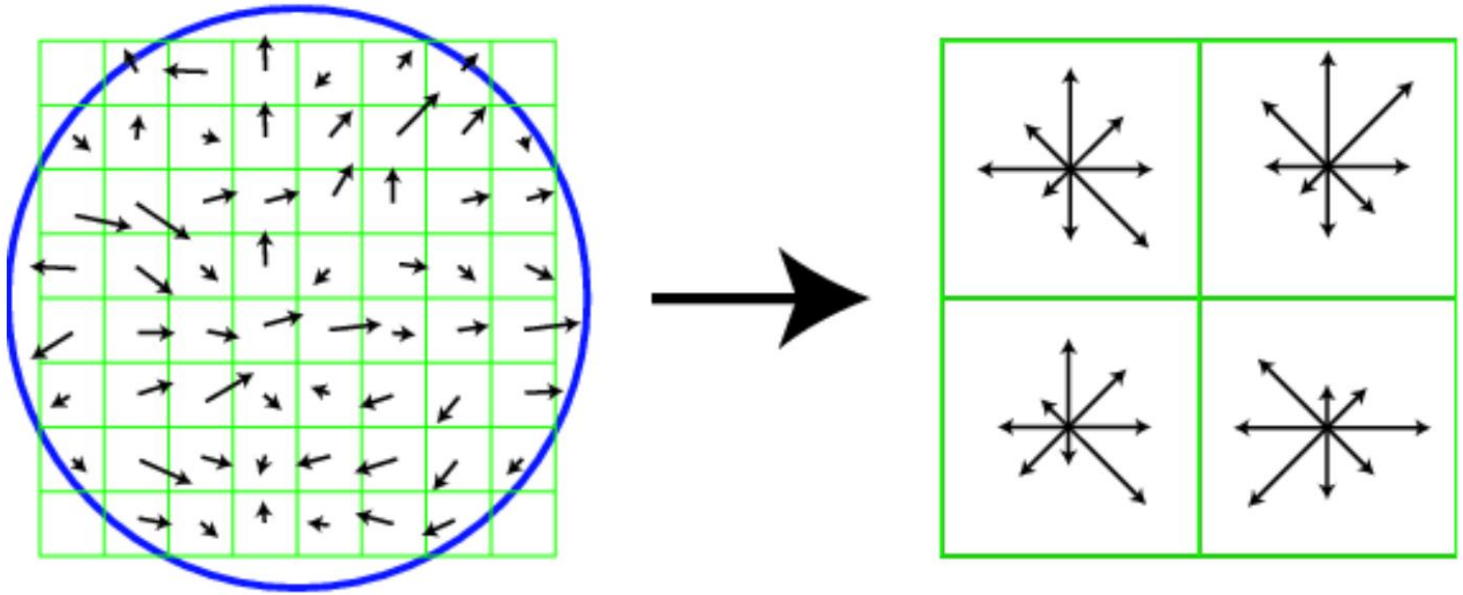


Image gradients

Keypoint descriptor

(d). The matrices in the left column are the output of applying Gaussian filters with different standard derivations for a single octave in the SIFT detection algorithm. There is **a single keypoint.** Your job is to find it based on extrema in scale-space. (the point in the middle scale is considered)

To find the keypoint, you first need to **build the Difference of Gaussian Images** in the scale-space. The key points are found at the locations of **extrema** in scale-space.

Gaussian Filtered Images

25	20	20	16
25	30	19	16
19	17	19	16
13	13	14	14

24	18	20	14
25	32	19	15
18	16	19	14
12	12	13	13

22	15	20	12
24	35	18	14
16	16	18	13
11	11	11	12

20	10	20	8
20	37	10	8
16	8	20	10
12	10	9	10

Gaussian Filtered Images

25	20	20	16
25	30	19	16
19	17	19	16
13	13	14	14

24	18	20	14
25	32	19	15
18	16	19	14
12	12	13	13

22	15	20	12
24	35	18	14
16	16	18	13
11	11	11	12

20	10	20	8
20	37	10	8
16	8	20	10
12	10	9	10

Difference of Gaussian Images

1	2	0	2
0	-2	0	1
1	1	0	2
1	1	1	1

scale = 1

2	3	0	2
1	-3	1	1
2	0	1	1
1	1	2	1

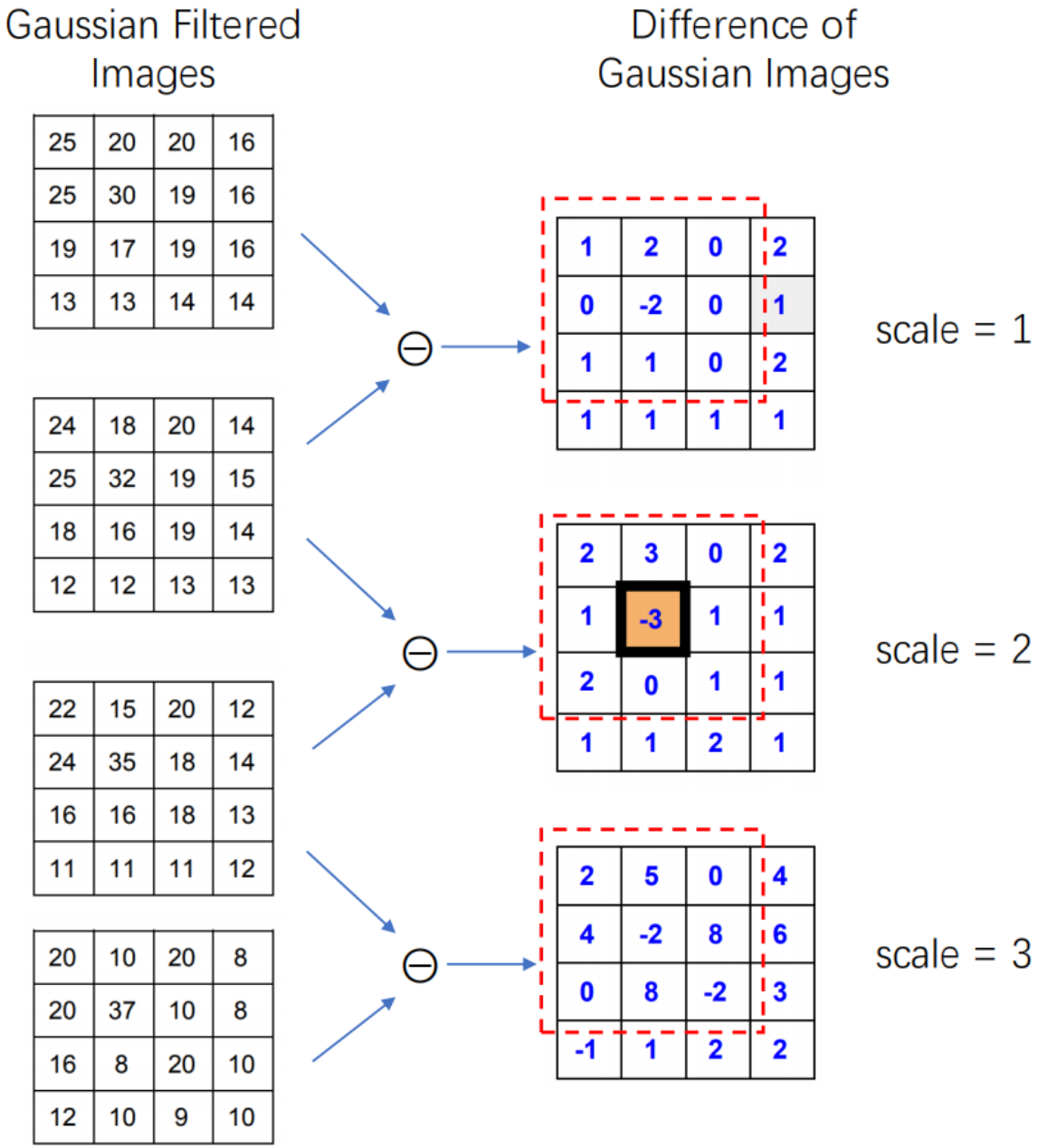
scale = 2

2	5	0	4
4	-2	8	6
0	8	-2	3
-1	1	2	2

scale = 3

The keypoint is located at x=1, y=1, scale=2, Since

Compared to neighbors at scale 1
-3 < 1,2,0,0,-2,0,1,1,0

Compared to neighbors in the same scale (scale 2)
-3 < 2,3,0,1, 1, 2,0,1

Compared to neighbors at scale 3
-3 < 2,5,0,4,-2,8,0,8,-2